

C-SPINE Revisited

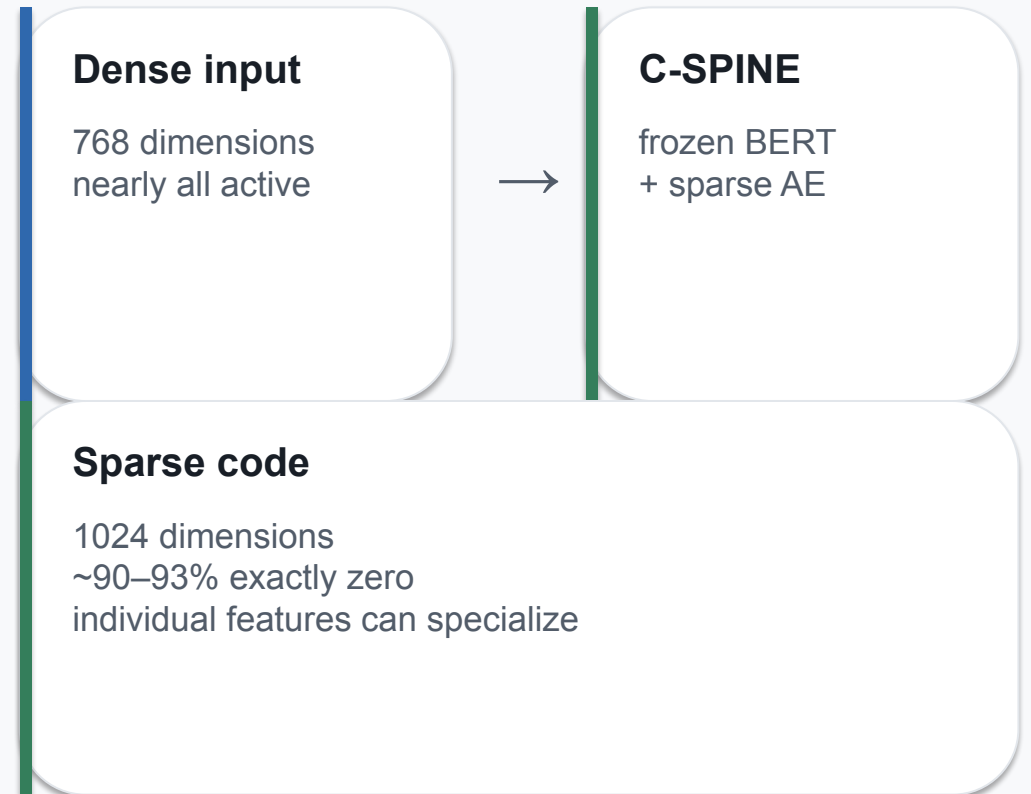
A Rigorous Replication, Audit, and Extension of
Sparse Interpretable Sentence Embeddings

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AI Academy Deep Learning | Final Project

The Problem: Dense Embeddings Are a Black Box

- BERT produces 768-dimensional contextual sentence vectors.
- Almost every coordinate is nonzero, so individual dimensions are hard to assign a meaning to.
- This opacity matters in high-stakes NLP: a useful representation should also be inspectable.
- C-SPINE asks: can we preserve downstream information while forcing most latent dimensions to zero?



From SPINE to C-SPINE

Sparse coding

Olshausen & Field: overcomplete sparse bases can represent signals efficiently and reduce redundancy.

SPINE

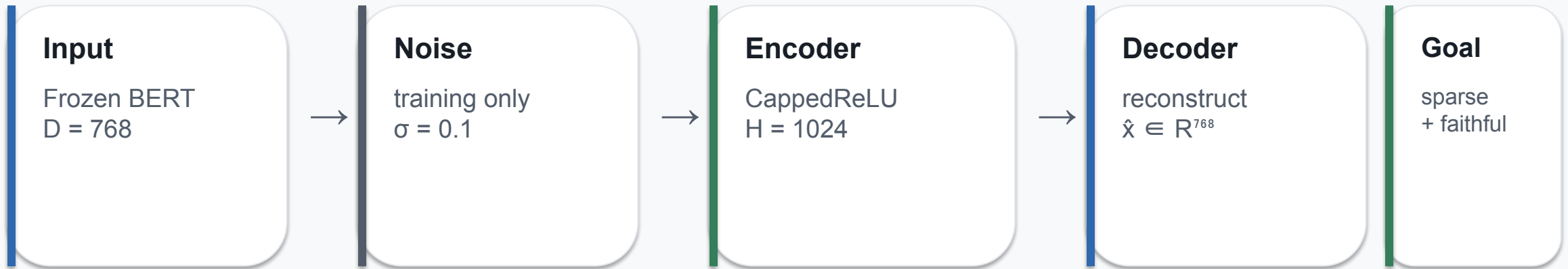
Subramanian et al.: denoising sparse autoencoder + CappedReLU + sparsity objective on static Word2Vec/GloVe embeddings.

C-SPINE

Ports the SPINE recipe to contextual BERT sentence embeddings — richer, context-dependent, and higher-variance.

This revision is a replication + audit + extension, not a new sparse-coding method.

Method: Architecture



- CappedReLU(v) = $\min(\max(0, v), 1)$: non-negative and bounded.
- Overcomplete $H > D$ gives the dictionary room to spread signal across sparse atoms.
- BERT is frozen; embeddings are extracted and cached once.
- The revision makes input normalization explicit via a configurable flag.

Method: Three-Part Loss

$$L = \text{reconstruction} + \lambda_1 \cdot \text{ASL} - \lambda_2 \cdot \text{PSL}$$

Reconstruction (MSE)

Keeps the sparse code informative enough to rebuild the original BERT vector.

ASL — Average Sparsity

Pushes mean activation of each hidden unit toward zero. This is the binding sparsity cost.

PSL — Peak Sparsity

Rewards a strong peak activation so units are not permanently dead.

ASL and PSL create a tension: features are expensive to use on average, but should still be used.

Full run: $\lambda_1 = \lambda_2 = 1.0$ • Adam • lr 10^{-3} • batch 64 • 30 epochs

What Changed in the Revisited Study?

1. Audit

We compared manuscript and shipped implementation across the pipeline.

Two material discrepancies emerged:

- paper claims L2-unit inputs; code does not normalize
- reporting split was reused for checkpoint/model selection

The new study restores a true validation split and locked test.

2. Stronger evaluation

Five evaluation domains, three seeds, a four-level sparsity sweep, mechanism/loss ablations, PCA and random-projection controls, feature-level metrics, and 25 ordered transfer directions.

3. Honest negatives

The extension tests where C-SPINE does not win:

- scientific topics are difficult
- feature identities are not stable across seeds
- sparse neighborhoods only partially match dense neighborhoods
- PCA often matches or beats sparse codes

Experimental Setup: Five Domains, Locked TEST

SST-2

sentiment • TEST 872

0.857 → 0.815

-4.1 pp

dense → sparse

AG News

4-way topics • TEST 7,600

0.903 → 0.875

-2.8 pp

dense → sparse

IMDB

long-review sentiment • TEST
25,000

0.814 → 0.749

-6.5 pp

dense → sparse

TREC

question classes • TEST 500

0.878 → 0.858

-2.0 pp

dense → sparse

arXiv-11

scientific topics • TEST 2,500

0.684 → 0.577

-10.7 pp

dense → sparse

Protocol

HF-train → stratified 90/10
train/VAL

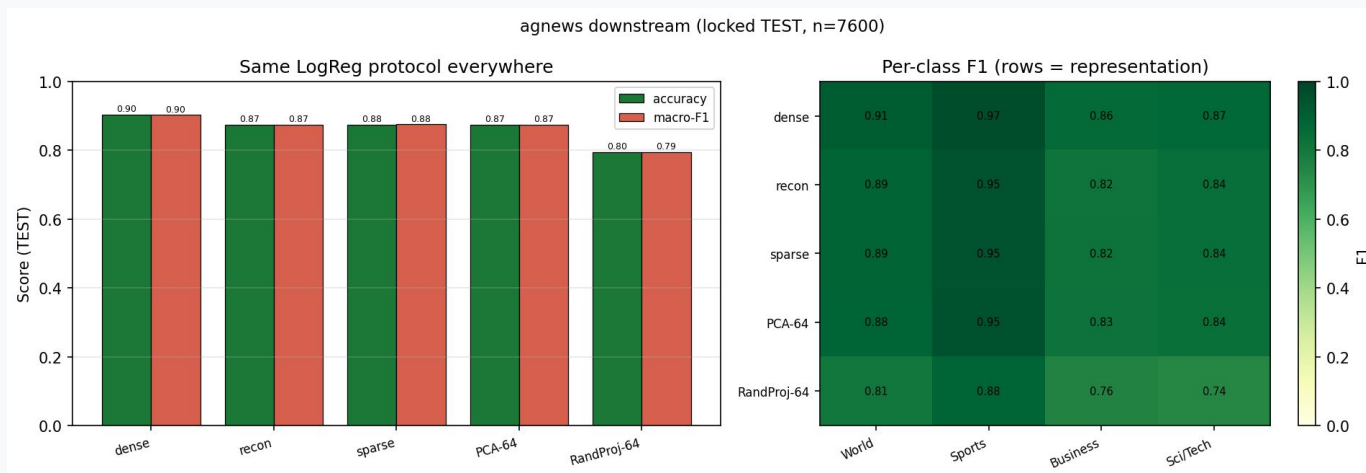
Best checkpoint by VAL
MSE only

Locked TEST touched
once

Logistic-regression linear
probes

Seeds 42 / 43 / 44

Results: Downstream Preservation



0.903

AG dense

0.875

AG sparse

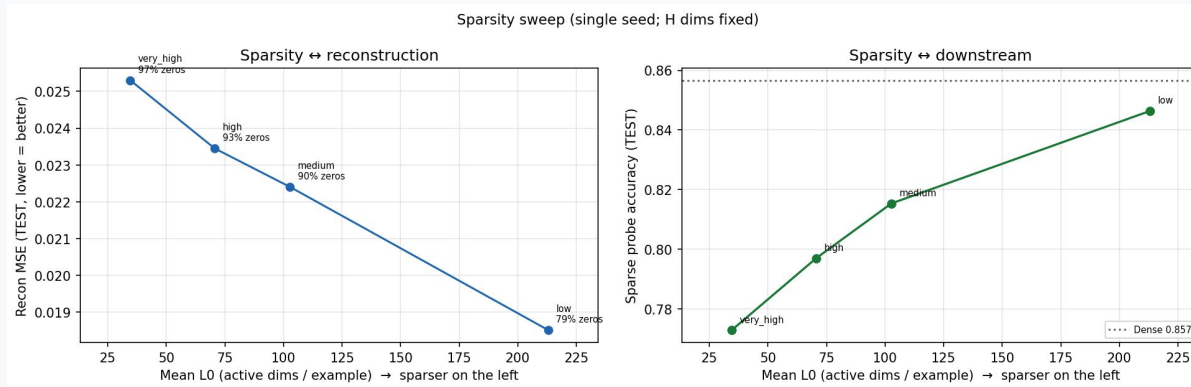
92.6%

sparsity

- AG News closely replicates the legacy headline.
- SST-2: 0.857 $\rightarrow$ 0.815 at 90.0% sparsity.
- IMDB: 0.814 $\rightarrow$ 0.749; TREC: 0.878 $\rightarrow$ 0.858.
- arXiv-11 is the failure case: 0.684 $\rightarrow$ 0.577.

**Across domains, strong compression is feasible
— but the amount of information retained
depends heavily on task structure.**

Results: Reconstruction & Sparsity Trade-off



SST-2 sweep

79.2% → 96.6% zeros as λ ASL increases 0.1 → 8.

Accuracy cost

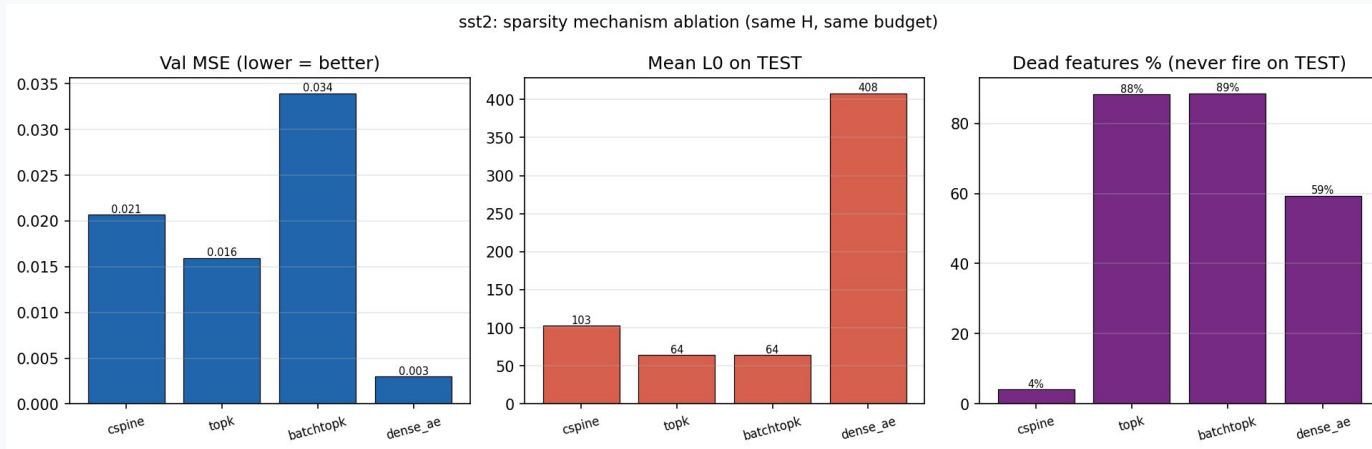
Sparse accuracy falls 0.846 → 0.773: a monotonic 7.3 pp decline.

No free plateau

MSE rises 0.0185 → 0.0253 while dead features rise 0.7% → 31.7%.

More sparsity is not free. The revision weakens the original “flat plateau” hypothesis.

Ablations: What Actually Makes C-SPINE Different?



TopK

Better VAL MSE (0.0159 vs 0.0207), but 88.3% of dictionary atoms are dead.

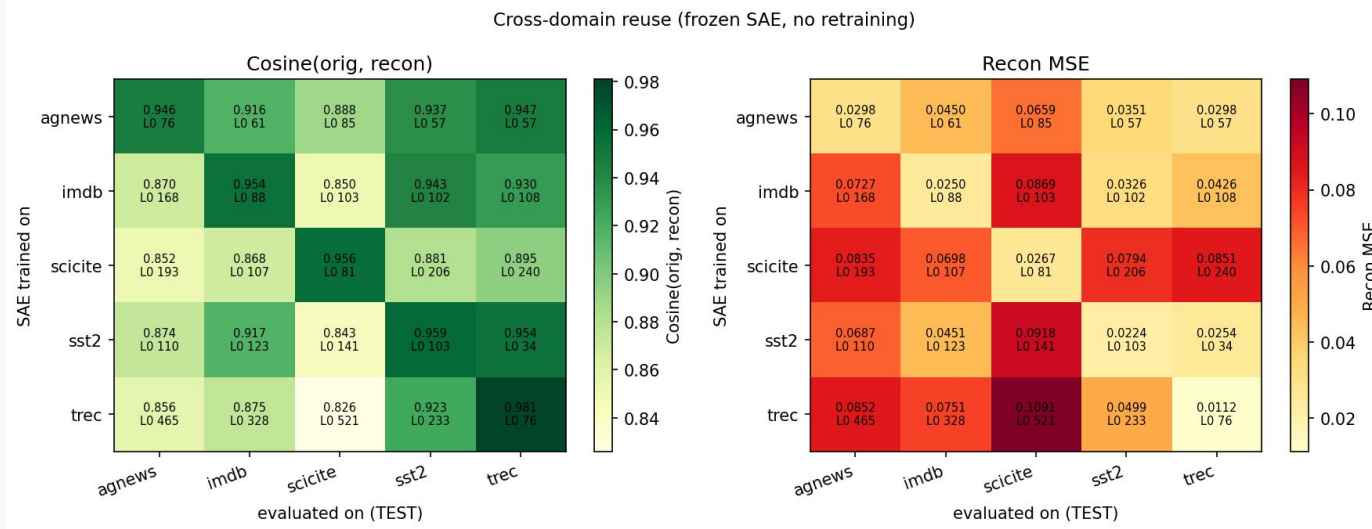
Dense AE

Best reconstruction (0.0030), but no learned sparsity; 59.4% dead ReLUs.

Key finding

C-SPINE's distinctive benefit is dictionary utilization: 4.1% dead and 72.0% utilization.

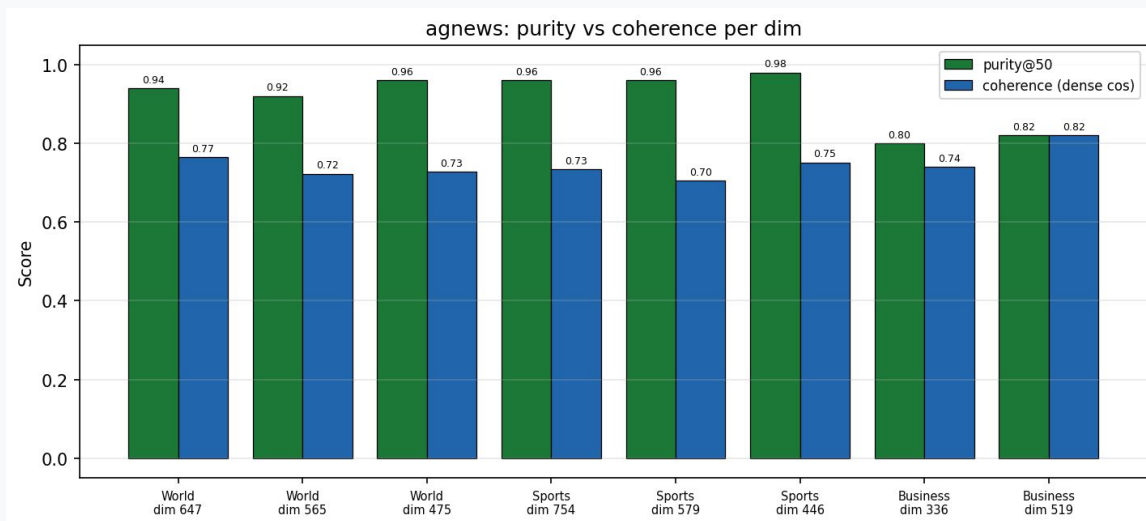
Transfer: Sparse Structure Is Domain-Specific



- 25 ordered train→test directions, with no retraining.
- In-domain cosine: 0.946–0.981.
- Off-diagonal MSE is often 1.5–4× worse.
- Best transfer: SST-2 ↔ IMDB sentiment.
- TREC dictionary over-fires on other corpora; other dictionaries often under-fire on TREC.

Compression transfers more readily than the exact learned geometry.

Interpretability: Some Features Really Do Specialize



AG News

Median purity@50 = 0.84
Median specialization = 0.89

12/12 showcased dims
majority-correct

SST-2

Median purity@50 = 0.60
Median specialization = 0.32

Sentiment is distributed

Example feature

AG dim 446 → Sports
purity 0.98 • specialization 0.973
fires on game recaps

Important distinction:
discriminative ≠
correlated ≠ coherent ≠
fully interpretable.

No labels are used to train the sparse autoencoder.

Six Hypotheses: Evidence, Not Just Headlines

H1	Mixed / partial	Strong compression works; scientific topics lose too much signal.
H2	Weakly supported / largely rejected	No broad flat accuracy plateau across sparsity levels.
H3	Unsupported	No coherence advantage over dense AE was demonstrated.
H4	Rejected	Mean matched feature cosine ≈ 0.305 across seeds.
H5	Mixed	In-domain utility survives; cross-domain geometry is brittle.
H6	Not supported	Sparse ties PCA-64 only on AG News; PCA leads elsewhere.

The strongest scientific result is narrower than the original claim: C-SPINE works, but its advantages are conditional.

Limitations & Next Steps

Limitations

- Normalization ablation remains queued.
- Interpretability is still not a human-annotator study.
- Feature identities are unstable across seeds.
- Scientific-topic generalization is weak.
- MiniLM / DistilBERT results are preliminary.
- No general beyond-compression advantage over PCA was established.

Next experiments

- Systematic σ / λ_1 / λ_2 search
- Dynamic Top-K / BatchTopK variants
- Larger encoders: BERT-large, RoBERTa-large
- End-to-end encoder + SAE training
- Formal human interpretability evaluation

Research principle

Keep the locked test untouched.

Report negative results.

Separate:
compression,
fidelity,
utilization,
interpretability,
and transfer.

This turns a promising idea into an auditable experiment.

Conclusion

90–93%

zeros in main codes

5

evaluation domains

25

transfer directions

0.84

AG median purity@50

0.305

seed-matched cosine

- C-SPINE can turn dense BERT sentence vectors into highly sparse, bounded codes while retaining substantial downstream signal.
- The strongest result is category-selective structure in AG News, emerging without label supervision.
- The revision also shows clear limits: no broad sparsity plateau, unstable feature identities, brittle transfer, and no general advantage over PCA at the tested operating point.
- The contribution is therefore a rigorous, reproducible characterization of where sparse contextual representations help — and where they do not.

C-SPINE: toward NLP representations that are both powerful and understandable.